

EC 441 Midterm Exam 1 — Solution Set

Spring 2026

Point Summary

Part I	Multiple Choice (10 questions $\times$ 2 pts)	20 pts
Part II	Short Answer (5 questions $\times$ 4 pts)	20 pts
Part III	Store-and-Forward Cubicle Maze	20 pts
Part IV	Error Control Coding & CRC	20 pts
Part V	Modulation, Entropy, and Noise	20 pts
Total		100 pts

Part I — Multiple Choice

[20 pts, 2 pts each]

Q1. Which delay type is most likely to vary unpredictably with traffic load?

Answer: D — **Queuing delay.** Queuing delay depends on buffer occupancy, which fluctuates with the instantaneous traffic load. Transmission and propagation delays are fixed by link capacity and distance; processing delay is roughly constant.

Q2. Minimum frame size in CSMA/CD exists so that:

Answer: B — **A station is still transmitting when the collision signal returns, enabling detection.** The minimum frame size ensures that the transmission time $t_t \geq 2\tau$ (the round-trip propagation time), so that a transmitter has not yet finished when it hears the collision. Without this guarantee, a short frame could be fully sent before the collision signal returns, making detection impossible.

Q3. Shannon entropy is best described as:

Answer: C — **The minimum average number of bits needed to represent one symbol losslessly.** This is the statement of Shannon's source coding theorem (first theorem). It is *not* the channel capacity (that is the Shannon–Hartley result for noisy channels).

Q4. In the TCP/IP stack, the Link layer is responsible for:

Answer: C — **Transferring frames between directly connected nodes on the same segment.** The Link layer handles node-to-node delivery using MAC addresses and operates within a single network segment. Routing across networks (A) is the Network layer; end-to-end reliable delivery (B) is the Transport layer; bit-to-signal conversion (D) is the Physical layer.

Q5. Non-zero CRC remainder: receiver

Answer: D — **Silently discards the frame.** Ethernet (IEEE 802.3) has no negative acknowledgment (NAK) mechanism at Layer 2. Error *recovery* (retransmission) is the responsibility of higher-layer protocols such as TCP.

Q6. As E_b/N_0 increases, $P_b = Q\left(\sqrt{2E_b/N_0}\right)$:

Answer: C — **Decreases — the argument of Q grows, driving Q toward zero.** $Q(x)$ is the tail probability of a standard Gaussian: $Q(x) \rightarrow 0$ as $x \rightarrow \infty$. Increasing E_b/N_0 increases the argument $\sqrt{2E_b/N_0}$, which pushes the decision threshold further into the noise tail. This produces the characteristic “waterfall” curve — BER falls steeply with increasing E_b/N_0 . Distractor D is a common misconception: the curve falls (not rises) steeply, hence “waterfall.”

Q7. An Ethernet switch learns ports by:

Answer: C — **Recording the source MAC address and arrival port of every incoming frame.** This is the self-learning (backward learning) algorithm. No explicit registration or ARP polling is required; the switch builds its forwarding table passively from traffic.

Q8. WiFi cannot use collision detection because:

Answer: C — **A transmitting node cannot listen while transmitting; it cannot hear its own collision.** A wireless transmitter's own signal completely overwhelms any incoming signal during transmission (the transmit power is orders of magnitude larger than the received signal). This is a fundamental hardware constraint, not a rate or regulatory limitation.

Q9. After the 3rd consecutive collision in binary exponential backoff, the station chooses from how many slot times?

Answer: C — **8** (2^3). After the k -th collision the station draws uniformly from $\{0, 1, \dots, 2^{\min(k, 10)} - 1\}$. For $k = 3$: $2^3 = 8$ possible values, i.e. the range $[0, 7]$ slot times. The window doubles with each successive collision (up to a cap of 1024), spreading retransmissions to reduce the probability of repeated collisions.

Q10. Manchester encoding is used in 10 Mb/s Ethernet rather than NRZ because:

Answer: B — **The guaranteed mid-bit transition enables clock recovery at the receiver.** Manchester encodes each bit as a transition (low→high for 1, high→low for 0), so there is always a transition at the midpoint of every bit period. The receiver can use these transitions to synchronize its clock continuously — no separate clock line is needed. NRZ has no guaranteed transitions during long runs of identical bits, causing clock drift. The cost is doubled bandwidth (one Manchester symbol occupies twice the bandwidth of one NRZ bit).

Part II — Short Answer

[20 pts, 4 pts each]

SA1. [4 pts] 500 KB file, $R = 10$ Mb/s, $d = 2000$ km, $v = 2 \times 10^8$ m/s.

(a) **Transmission delay:**

$$t_t = \frac{L}{R} = \frac{500 \times 10^3 \times 8 \text{ bits}}{10 \times 10^6 \text{ b/s}} = \frac{4 \times 10^6}{10^7} = \boxed{0.4 \text{ s}}$$

(b) **Propagation delay:**

$$t_p = \frac{d}{v} = \frac{2 \times 10^6 \text{ m}}{2 \times 10^8 \text{ m/s}} = \boxed{0.01 \text{ s} = 10 \text{ ms}}$$

SA2. [4 pts] $B = 1$ MHz, $\text{SNR} = 31$ (linear). Modem claims 4.5 Mb/s.

Shannon capacity:

$$C = B \log_2(1 + \text{SNR}) = 10^6 \times \log_2(32) = 10^6 \times 5 = \boxed{5 \text{ Mb/s}}$$

Since $4.5 \text{ Mb/s} < 5 \text{ Mb/s} = C$, the claim **is theoretically possible** (not forbidden by Shannon's theorem).

SA3. [4 pts] Slotted ALOHA collision fraction, $G = 0.5$, $e^{-0.5} \approx 0.6$.

A collision occurs when $k \geq 2$ frames arrive in the same slot.

$$P(\text{collision}) = 1 - P(k = 0) - P(k = 1)$$

Compute each term with $G = 0.5$:

$$P(k=0) = \frac{0.5^0 e^{-0.5}}{0!} = e^{-0.5} \approx 0.6$$

$$P(k=1) = \frac{0.5^1 e^{-0.5}}{1!} = 0.5 e^{-0.5} \approx 0.3$$

$$P(\text{collision}) = 1 - 0.6 - 0.3 = \boxed{0.1 = 10\%}$$

SA4. [4 pts] ARP scenario (A: AA:AA:AA:AA:AA:AA, B: BB:BB:BB:BB:BB:BB).

(a) ARP request destination MAC:

$$\boxed{\text{FF:FF:FF:FF:FF:FF}} \quad (\text{Ethernet broadcast})$$

The ARP request must reach all hosts on the segment because A does not yet know B's MAC.

(b) After B replies, A sends the IP datagram encapsulated in an Ethernet frame:

- Ethernet source MAC: $\boxed{\text{AA:AA:AA:AA:AA:AA}}$
- Ethernet destination MAC: $\boxed{\text{BB:BB:BB:BB:BB:BB}}$

SA5. [4 pts] $P_t = 20$ dBm, cable/connector loss = 3 dB, path loss = 100 dB, $P_{\min} = -85$ dBm.

(a) **Received power:**

$$P_r = P_t - \text{losses} = 20 - 3 - 100 = \boxed{-83 \text{ dBm}}$$

(b) **Link margin:**

$$\text{Margin} = P_r - P_{\min} = -83 - (-85) = \boxed{+2 \text{ dB}}$$

The link margin is positive, so the link is **viable** (barely — only 2 dB of margin).

Part 3: Store-and-Forward in the Cubicle Maze [20 pts]

(a) [10 pts] **Message format design.**

The system has $5 \times 5 = 25$ cubicles, addressed as (r, c) with $r, c \in \{0, \dots, 4\}$.

Required fields:

1. **Destination address** — needed to route the message to the correct cubicle. With 25 cubicles, $\lceil \log_2 25 \rceil = 5$ bits suffice for a flat address. Alternatively, two coordinate fields of $\lceil \log_2 5 \rceil = 3$ bits each = 6 bits.
2. **Source address** — needed so the recipient knows who sent the message (e.g., to compose a reply). Same size as destination: 5 bits (flat) or 6 bits (coordinate).
3. **Payload** — given as 8 bits.

Total message length (using 5-bit flat addresses):

$$L = 5 + 5 + 8 = \boxed{18 \text{ bits}}$$

Using 3-bit coordinate pairs: $L = 6 + 6 + 8 = 20$ bits.

(b) [10 pts] **End-to-end delay.** (Using $L = 18$ bits, $R = 2$ b/s, $\tau = 0.2$ s.)

(i) **Number of hops:** Path $(0,0) \rightarrow (0,1) \rightarrow (0,2) \rightarrow (0,3) \rightarrow (0,4)$ has 5 nodes and 4 hops.

(ii) **Delays per hop:**

$$t_t = \frac{L}{R} = \frac{18}{2} = 9 \text{ s per hop}$$

$$\tau = 0.2 \text{ s per hop}$$

(iii) **Total end-to-end delay (store-and-forward):**

In store-and-forward, each intermediate node must receive the full message before retransmitting it. The total delay for N hops is:

$$T_{total} = N \times (t_t + \tau) = 4 \times (9 + 0.2) = 4 \times 9.2 = \boxed{36.8 \text{ s}}$$

(iv) **Time of first-bit arrival at $(0,4)$:**

Node $(0,3)$ begins transmitting at the moment it has received the full message:

$$t_{\text{start at } (0,3)} = 3 \times (t_t + \tau) = 3 \times 9.2 = 27.6 \text{ s}$$

The first bit of the message propagates from $(0,3)$ to $(0,4)$ in time τ :

$$t_{\text{first bit at } (0,4)} = 27.6 + \tau = 27.6 + 0.2 = \boxed{27.8 \text{ s}}$$

Comparison:

$$T_{total} - t_{\text{first bit}} = 36.8 - 27.8 = 9 \text{ s} = t_t$$

The difference equals *one transmission time* t_t . This is the **serialization delay** of the last hop: the time required to clock all 18 bits from node $(0,3)$ into the channel. The first bit arrives at $(0,4)$ after 27.8 s; the last bit arrives 9 s later, at 36.8 s, marking the completion of delivery.

Part 4: Error Control Coding & CRC [20 pts]

Generator: $G(x) = x^3 + x + 1$ (bit string: 1011); $k = 3$, $r = 3$, $n = 6$.

(a) [6 pts] Find the codeword for message 101.

Step 1: Append $r = 3$ zeros to the message.

$$101 \rightarrow 101000$$

Step 2: Divide 101000 by 1011 using XOR (mod-2) long division.

$$\begin{array}{r}
 101000 \div 1011 \\
 \text{Align: } 1010 \\
 \oplus 1011 \\
 \hline
 0001 \quad \Rightarrow \text{bring down } 0 : 00010 \\
 \text{Align: } 0010 \\
 (\text{MSB} = 0, \text{ no subtraction; bring down } 0): 000100 \\
 \text{Align: } 0100 \\
 (\text{MSB} = 0, \text{ no more bits.}) \\
 \text{Remainder} = 100
 \end{array}$$

Step 3: Codeword = message || remainder:

$$\boxed{101100}$$

Verification: $101100 \div 1011$ yields remainder 000. ✓

Complete codeword table:

Message	Codeword
000	000000
001	001011
010	010110
011	011101
100	100111
101	101100
110	110001
111	111010

(b) [8 pts] Determine d_{min} .

For a *linear* code, d_{min} equals the minimum Hamming weight of any **non-zero** codeword (since the XOR of any two codewords is also a codeword, the minimum pairwise distance equals the minimum weight).

Codeword	Hamming weight
000000	0 (zero codeword, excluded)
001011	3
010110	3
011101	4
100111	4
101100	3
110001	3
111010	4

$$d_{min} = 3$$

Error detection capability: detects up to $d_{min} - 1 = 2$ bit errors.

Error correction capability: corrects up to $\left\lfloor \frac{d_{min} - 1}{2} \right\rfloor = 1$ bit error.

(c) [6 pts] Received word: 001111.

Compute Hamming distances to all codewords (equivalently, Hamming weight of XOR):

Codeword	XOR with 001111	Distance
000000	001111	4
001011	000100	1 ← minimum
010110	011001	3
011101	010010	2
100111	101000	2
101100	100011	3
110001	111110	5
111010	110101	4

Nearest codeword: 001011 (distance 1). **Decoded message:** 001.

Does the receiver make an error? No. Assuming a single-bit channel error occurred (bit 3 from the right was flipped), the transmitted codeword was 001011 and minimum-distance decoding correctly recovers it. Since $d_{min} = 3$ and only 1 error occurred, single-error correction is guaranteed to succeed.

Under what circumstances would this decoder make a mistake? When **2 or more bit errors** occur. Specifically, if 2 errors shift the received word closer to a *wrong* codeword than to the transmitted codeword, the

decoder will output the wrong message (a miscorrection). For example, if 011101 was transmitted and 2 bits were corrupted to produce 001111, the decoder would output 001011 — an undetected miscorrection.

Part 5: Modulation, Entropy, and Noise [20 pts]

5-QAM: $s_0 = (0, 0)$ with $p = 1/2$; corners $(\pm A, \pm A)$ each with $p = 1/8$.

4-QAM: corners $(\pm A, \pm A)$ each with $p = 1/4$.

(a) [8 pts] Compute $H(X)$ for both constellations.

5-QAM:

$$H_5 = -\frac{1}{2} \log_2 \frac{1}{2} - 4 \cdot \frac{1}{8} \log_2 \frac{1}{8} = \frac{1}{2} \cdot 1 + 4 \cdot \frac{1}{8} \cdot 3 = \frac{1}{2} + \frac{3}{2} = \boxed{2 \text{ bits/symbol}}$$

4-QAM:

$$H_4 = -4 \cdot \frac{1}{4} \log_2 \frac{1}{4} = 4 \cdot \frac{1}{4} \cdot 2 = \boxed{2 \text{ bits/symbol}}$$

Both constellations have **identical entropy** of 2 bits/symbol. They carry the same information per symbol.

(b) [6 pts] Determine d_{min} for each constellation (at the same amplitude A).

4-QAM: The four points are at $(\pm A, \pm A)$. Adjacent corners (e.g., (A, A) and $(-A, A)$) are the nearest neighbors:

$$d_{min}^{(4)} = \sqrt{(2A)^2 + 0^2} = \boxed{2A}$$

5-QAM: The nearest pair is the center $(0, 0)$ and any corner (A, A) :

$$d_{min}^{(5)} = \sqrt{A^2 + A^2} = \boxed{A\sqrt{2} \approx 1.414A}$$

At the same amplitude A : $A\sqrt{2} < 2A$, so **4-QAM has greater noise immunity**.

(c) [6 pts] Average power and equal-power comparison.

5-QAM average power:

$$P_{avg}^{(5)} = \frac{1}{2}(0^2 + 0^2) + 4 \cdot \frac{1}{8}(A^2 + A^2) = 0 + \frac{1}{2} \cdot 2A^2 = \boxed{A^2}$$

4-QAM average power:

$$P_{avg}^{(4)} = 4 \cdot \frac{1}{4}(A^2 + A^2) = 2A^2$$

Equal-power scaling: Set both to average power P .

5-QAM: $A^2 = P \Rightarrow A = \sqrt{P}$, so $d_{min}^{(5)} = A\sqrt{2} = \sqrt{2P}$.

4-QAM: $2A^2 = P \Rightarrow A = \sqrt{P/2}$, so $d_{min}^{(4)} = 2A = 2\sqrt{P/2} = \sqrt{2P}$.

At equal average power:

$$d_{min}^{(5)} = d_{min}^{(4)} = \sqrt{2P}$$

Is 5-QAM a better design? No. At equal average power and equal entropy (2 bits/symbol), both constellations achieve *identical* minimum distance $\sqrt{2P}$. The center point in 5-QAM reduces average power (good), but reduces the minimum distance by exactly the same factor, leaving noise immunity unchanged. 5-QAM offers no advantage over 4-QAM on an AWGN channel.